

Dobrushin's uniqueness condition for the Wilson action: a two-line strong-coupling gap at $g^2 > 444$, and the block criterion that is the finite verification

The finite-volume criteria of the finite-verification note have a single-site ancestor, Dobrushin's uniqueness condition (Theory Probab. Appl. 13 (1968) 197), and for the Wilson action it can be checked by hand. The conditional distribution of one link given all the others is

$$d\mu_\ell(U | \omega) \propto \exp \left[\frac{\beta_W}{N} \text{Re tr} (U M_\ell(\omega)) \right] d\mu(U), \quad M_\ell = \sum_{p \ni \ell} S_p,$$

the sum of the six staples of ℓ in four dimensions. Changing one neighbouring link changes one staple, and the resulting total-variation distance between the two conditional distributions is at most $e^{4\beta_W} - 1$, independently of N . Each link has 18 neighbours that share a plaquette with it, so Dobrushin's coefficient is

$$\alpha = \sup_\ell \sum_{\ell'} \rho_{\ell\ell'} \leq 18(e^{4\beta_W} - 1),$$

and the condition $\alpha < 1$ holds for $\beta_W < \frac{1}{4} \log \frac{19}{18} = 0.01352$, that is

$$g^2 > 148 N = 444 \quad (SU(3)).$$

Dobrushin's theorem gives a unique Gibbs state, Künisch's theorem (Commun. Math. Phys. 84 (1982) 207) gives exponential decay of all truncated correlations at rate $\log(1/\alpha)$ per lattice unit, and link reflection positivity turns the rate into a gap:

$$\Delta_W \geq \frac{\hbar c}{a} \log \frac{1}{18(e^{24/g^2} - 1)} \simeq \frac{\hbar c}{a} \log \frac{g^2}{432} \quad (g^2 > 444),$$

uniformly in the volume. This is a better threshold than the polymer expansion's 1056 and a smaller rate than its $4 \log(g^2/1056)$. The block version of the same condition, the Dobrushin–Shlosman constructive criterion (in *Statistical Physics and Dynamical Systems*, Birkhäuser 1985, 347), replaces the single link by a box V of links and is expected to hold deeper into the intermediate region as V grows; it is the quantity a finite verification would compute, and Section 4 states it as a specification. All references at metadata level; constants explicit; nothing promoted.

1. The one-link conditional distribution

Fix all links except ℓ . The plaquettes containing ℓ are six in four dimensions, and each contributes $\text{Re tr}(U_\ell S_p)$ with $S_p \in SU(N)$ the product of the other three links of p in the appropriate order. Hence the conditional density is $e^{f_\omega(U)}$ with $f_\omega(U) = (\beta_W/N) \text{Re tr}(U M_\ell)$, $M_\ell = \sum_{p \ni \ell} S_p$, normalized by $Z(\omega) = \int e^{f_\omega} d\mu$. Gauge invariance of the interaction is visible in the fact that a gauge transformation at an endpoint of ℓ rotates M_ℓ and the Haar measure absorbs it; nothing more is needed.

2. The Dobrushin coefficient

Lemma 1. For probability densities $p \propto e^f$, $q \propto e^{f'}$ with respect to the same measure, $\|p - q\|_{\text{TV}} \leq e^{\text{osc}(f-f')} - 1$.

Proof. $p/q = e^{f-f'} Z_{f'}/Z_f \leq e^{\sup(f-f')} e^{\sup(f'-f)} = e^{\text{osc}(f-f')}$, so $p - q \leq (e^{\text{osc}} - 1)q$ and $\|p - q\|_{\text{TV}} = \int (p - q)_+ \leq e^{\text{osc}} - 1$. $\square$

Lemma 2. If ω, ω' differ only at a link ℓ' sharing a plaquette with ℓ , then $\text{osc}_U(f_\omega - f_{\omega'}) \leq 4\beta_W$.

Proof. One staple changes, $M_\ell - M'_\ell = S_p - S'_p$ with $S_p, S'_p \in SU(N)$, so $|f_\omega(U) - f_{\omega'}(U)| = (\beta_W/N) |\text{Re tr}(U(S_p - S'_p))| \leq (\beta_W/N) \|S_p - S'_p\|_1 \leq (\beta_W/N) \cdot 2N$, the trace norm of a difference of two unitaries being at most $2N$; the oscillation is at most twice the supremum. $\square$

If ℓ' shares no plaquette with ℓ , the conditional distribution at ℓ does not depend on it. Each link lies in six plaquettes with three further links each, and the six plaquettes share no link besides ℓ , so there are exactly 18 links ℓ' with $\rho_{\ell\ell'} \neq 0$. Hence

$$\alpha = \sup_\ell \sum_{\ell'} \rho_{\ell\ell'} \leq 18(e^{4\beta_W} - 1),$$

and $\alpha < 1$ for $\beta_W < 0.01352$, i.e. $g^2 > 2N/0.01352 = 148N$.

3. From the condition to the gap

Dobrushin's theorem: $\alpha < 1$ implies a unique Gibbs state on $\mathbb{Z}^4$ and on every torus. The covariance estimate of Föllmer (J. Funct. Anal. 46 (1982) 387; Künsch, loc. cit., for the decay statements): under $\alpha < 1$, for local observables F, G ,

$$|\langle F; G \rangle| \leq \sum_{\ell \in \text{supp } F, \ell' \in \text{supp } G} \delta_\ell F [(1 - \rho)^{-1}]_{\ell\ell'} \delta_{\ell'} G,$$

with $\delta_\ell F$ the oscillation of F in the link ℓ , and since $\rho_{\ell\ell'}^n$ vanishes unless the link distance is at most n lattice units, $[(1 - \rho)^{-1}]_{\ell\ell'} \leq \alpha^{d(\ell, \ell')}/(1 - \alpha)$. Truncated correlations at time separation T therefore decay at least like $\alpha^{T/a}$, rate $m = \log(1/\alpha)/a$. The Wilson transfer matrix is positive by link reflection positivity, and for a local A orthogonal to the vacuum $\langle A, \mathcal{T}^n A \rangle \leq C_A \alpha^n$ forces its spectral measure to vanish above α , so on the cyclic subspace of local observables

$$\Delta_W = -\frac{\hbar c}{a} \log \|\mathcal{T}|_{\Omega^\perp}\| \geq \frac{\hbar c}{a} \log \frac{1}{\alpha} \geq \frac{\hbar c}{a} \log \frac{1}{18(e^{24/g^2} - 1)} \quad (SU(3)).$$

route	rigorous threshold ($SU(3)$)	gap bound
polymer expansion, all representations	$g^2 \geq 1056$	$(\hbar c/a) 4 \log(g^2/1056)$
Dobrushin single-site condition	$g^2 > 444$	$(\hbar c/a) \log(g^2/432)$

The two are complementary: the Dobrushin route enters earlier, the polymer route's rate is larger once both apply. For $U(1)$ the same computation gives $g^2 > 148$ with $N = 1$.

4. The block criterion, as a specification

The Dobrushin–Shlosman constructive criterion replaces the single link by a finite box V of links. Writing $\mu_V(\cdot|\omega)$ for the Gibbs measure in V with boundary condition ω and, for $x \in V$ and $y \notin V$,

$$\rho_V(x, y) = \sup_{\omega, \omega' \text{ differ only at } y} \|\mu_V(\cdot|\omega)|_x - \mu_V(\cdot|\omega')|_x\|_{\text{TV}},$$

the criterion is that for some finite V

$$\sum_{x \in V} \sum_{y \notin V} \rho_V(x, y) < |V|,$$

up to the normalization convention of the original, and it implies the same conclusions as the single-site condition, which is the case $V = \{\ell\}$. Its content is that the total influence of the boundary on the box, summed over the box, is less than the size of the box; in a mixing phase this holds once the side of V exceeds a few correlation lengths, which is why it can hold at couplings where the single-site condition fails. Dobrushin and Shlosman's complete analyticity (*Statistical Physics and Dynamical Systems*, Birkhäuser 1985, 371; J. Stat. Phys. 46 (1987) 983) is the statement that some such V exists, together with its equivalent forms.

What a verification computes. For a box V of side R (about $3R^4$ links) and a coupling β_W : for each boundary link y and each interior link x , a rigorous upper bound on $\rho_V(x, y)$, which is a total-variation distance between two marginals of Gibbs measures on $SU(3)^{|V|}$; then the double sum against $|V|$. Lemma 1 bounds each $\rho_V(x, y)$ by $e^{\text{osc}} - 1$ of the log-density difference of the two marginals, but for $|V| > 1$ that oscillation involves the integral over the other links of V and is what has to be bounded rigorously. The openness radius in the interaction norm $\sup_{\ell} \sum_{p \ni \ell} \|\Phi_p\|_{\infty}$ follows from the margin $|V| - \sum \rho_V$ by the same lemma.

Where it would have to hold. The numerical evidence places the $SU(3)$ crossover at $\beta_W \simeq 5\text{--}6.5$ with ξ/a of order 1 to 10; a verification at those couplings would need R of order 10–30, that is $10^4\text{--}10^6$ links, and rigorous bounds on the corresponding integrals. That is far beyond present practice, and this repository performs no numerics. The specification is nonetheless complete: the quantity, the boxes, the inequality, and the two lemmas that convert its margin into a gap and into an openness radius.

5. Consequence for STATE

The strong-coupling side now has a third, elementary proof with the best rigorous threshold for the Wilson transfer matrix, $g^2 > 444$ for $SU(3)$, and the finite verification of the intermediate region has a precise statement: the Dobrushin–Shlosman block criterion for the Wilson interaction, whose single-link case is proved here. The two numbers of the finite-verification note are unchanged, and the second of them, g_{DS}^2 , is exactly the smallest coupling at which the block criterion has been verified for some V .